Working code for the biomechzoo Ensembler

This is the plotting library associated with the biomechzoo toolbox. This means that the data inputted into these plotting functions need to be *zoo files* or structured as nested dictionaries.

```
# import statements
from src.ensembler import Ensembler
from src.plot_spec import PlotSpec
from src.renderers import (IndividualLinesRenderer,
                           MeanSDRenderer, CompositeRenderer,
                           ViolinRenderer, EventOverlayRenderer)
from src.helpers import ConditionSpec, ConditionSource
```

Variables

The following variables should be set a priori for it to work

1. **in_folder**: path to the root where the data is stored
2. **conditions**: Optional if there are multiple conditions.
3. **channels**: the channel names that you'd want to plot
4. **str_match**: list of regular expression that matches the subject names. if emtpy; provide a subject list
5. **subj_list**: list of strings containing the subject names. if empty; provide a str_match.
6. **rows**: number of subplot rows
7. **cols**: number of subplot cols.

`condition_spec = ConditionSpec(source = ConditionSource.FOLDER, conditions= ["Pre-surgery", "Post-surgery"])`

```
# Set up variables.
fld = "/Users/Werk/Documents/Postdoc-McGill/areve-mcgill/data/Round2/16-qc"

spec = ConditionSpec(
    source     = ConditionSource.CHANNEL,
    conditions = ["vicon", "areve", 'pig'],
    channel_map = {
        "vicon": "RS_abduction_vicon",
        "areve": "RS_abduction_areve",
        "pig" : "RS_abduction_pig",
    }
)
channels = ['RS_abduction']
str_match = [r"\b\d{3}[A-Z]{2}\b", r"\b\d{3}[A-Z]{3}\b"]
# subj_list = ["001CEJ"]
events = ["max"]
```

```
rows = 1
cols = 3
```

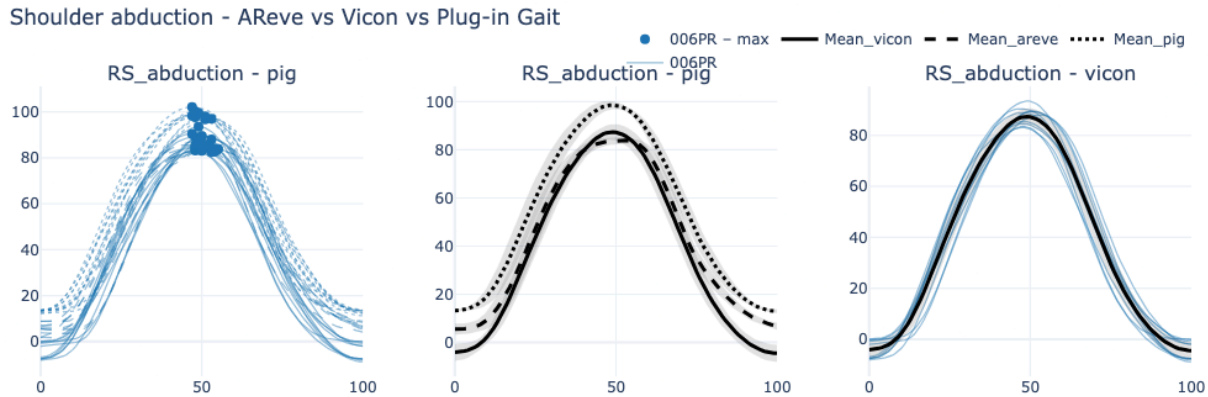
Using the ensembler

This is a collection of classes that work together to obtain clean and consistent plotting. Currently implemented figures are:

1. Individual line plots (`IndividualLinesRendered()`)
2. Average +/- standard deviation of curves (`MeanSDRenderer()`)

`CompositeRendered()` allows the uses to combine different renderers into a single subplot without needing extra lines of code. All renders are called in the `add_subplot` method from the Ensembler class alongside the channel name, row/col combination in which subplot to put it.

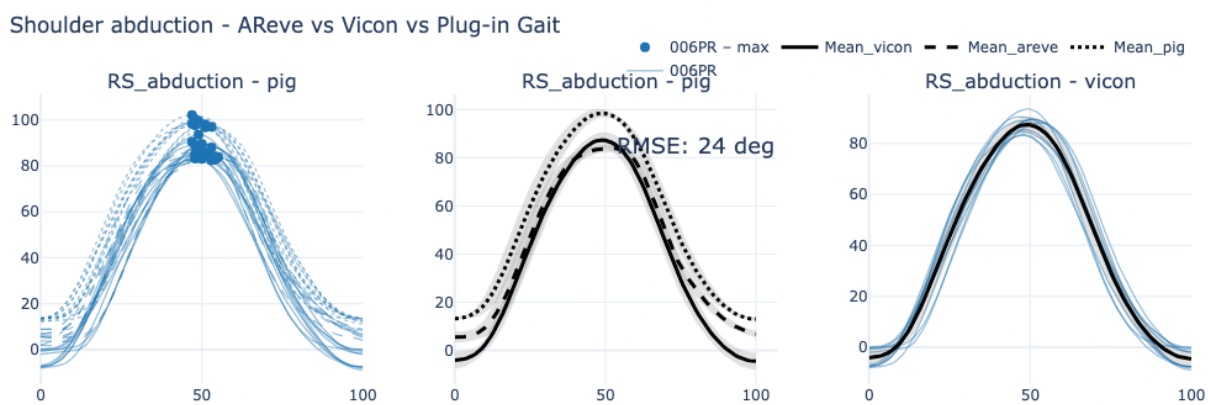
```
lines_and_mean = CompositeRenderer(IndividualLinesRenderer(), MeanSDRenderer())
lines_and_events = CompositeRenderer(IndividualLinesRenderer(), EventOverlayRenderer(

fig = (
    Ensembler(
        in_folder=fld, channels=channels,
        n_rows=rows, n_cols=cols,
        str_match=str_match, condition_spec=spec)
    .add_subplot(PlotSpec(
        'RS_abduction',   "vicon",
        row=1, col=1,
        renderer=lines_and_events, events=events))
    .add_subplot(PlotSpec(
        'RS_abduction', "areve",
        row=1, col=1,
        renderer=lines_and_events, events=events))
    .add_subplot(PlotSpec(
        'RS_abduction', "pig",
        row=1, col=1,
        renderer=lines_and_events,events=events))

    .add_subplot(PlotSpec(
        'RS_abduction', "vicon",
        row=1, col=2,
        renderer=MeanSDRenderer()))
    .add_subplot(PlotSpec(
        'RS_abduction', "areve",
        row=1, col=2,  renderer=MeanSDRenderer()))
    .add_subplot(PlotSpec(
        'RS_abduction', "pig",
        row=1, col=2,  renderer=MeanSDRenderer()))
    .add_subplot(PlotSpec(
        'RS_abduction', "vicon",
        row=1, col=3,  renderer=lines_and_mean))
    .build(title="Shoulder abduction - AReve vs Vicon vs Plug-in Gait")
)
fig.show()
```



Extras

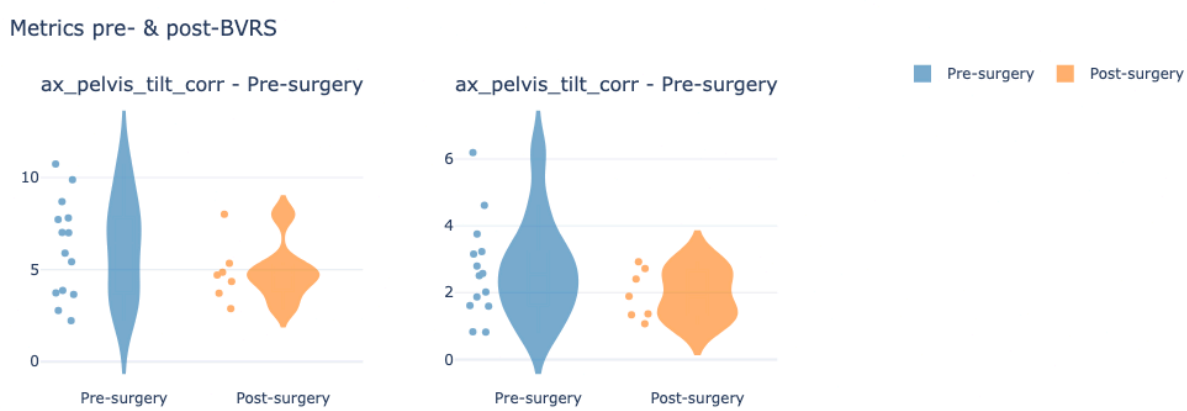
The returned fig from the ensembler allows the user all the functionality of plotly figures. e.g. add annotations to the plot

```
fig.add_annotation(x=80, y=80, showarrow=False,
            text="RMSE: 24 deg",
            yshift=10, font=dict(size=18), row=1,col=2)
```



Our first statistical plot

```
# Set up variables
fld = "/Users/Werk/Documents/Postdoc-McGill/breast-reduction/data/stats"
spec = ConditionSpec(
    source = ConditionSource.FOLDER,
    conditions = ["Pre-surgery", "Post-surgery"]
)

channels = ['ax_pelvis_tilt_corr']
str_match = [r"\b\d{3}[A-Z]{2}\b", r"\b\d{3}[A-Z]{3}\b"]
```

```
events = ['impact_peak_mean', 'loading_rate_mean']
rows = 1
cols = 3
```

```
fig = (
    Ensembler(
        in_folder=fld,  channels=channels,
        n_rows=rows,  n_cols=cols,
        str_match=str_match,
        condition_spec=spec,
        events=events)
    .add_subplot(PlotSpec(
        channel="ax_pelvis_tilt_corr",
        condition = "Pre-surgery", companions = ["Post-surgery"],
        row=1, col=1,
        renderer=ViolinRenderer(),
        events=['impact_peak_mean'],)
    )
    .add_subplot(PlotSpec(
        channel="ax_pelvis_tilt_corr",
        condition="Pre-surgery", companions=["Post-surgery"],
        row=1, col=2,
        renderer=ViolinRenderer(),
        events=['loading_rate_mean'])
    )
    .build(title="Metrics pre- & post-BVRS")
)
fig.show()
```